

Data Science-Driven System Design with LSTM Neural Networks: A COVID-19 Perspective

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Abstract—This paper emphasizes the significance of utilizing data science tools for system design research and their contribution to the field. It suggests selecting appropriate tools based on evaluations of functionality, scalability, and performance, aligning with system goals. A system design presented in this paper is based on data science tools, focusing on a three-layer architecture for an information system. A platform is developed to monitor and predict COVID-19 epidemiological data, addressing data quality issues through preprocessing, outlier detection, and imputation. The study employs advanced deep learning for prediction models, supported by robust validation methodologies. The resulting graphical interface facilitates effective visualization, aiding researchers in comprehending COVID-19 trends and making informed decisions. Overall, the research underscores the impact of data analytics, predictive modeling, and data-driven decision-making on complex phenomena and digital transformation.

Keywords—COVID-19, data science, deep learning, system design, graphical user interface

I. INTRODUCTION

In the pursuit of constructing a data-centric system, a comprehensive grasp of data categorizations and their origins is paramount. This understanding is pivotal as it dictates viable analytical methodologies, particularly within the context of machine learning, as highlighted by [1]. In alignment with the research conducted by authors Nandi and Sharma. [2], the spectrum of analytics encompasses descriptive, diagnostic, predictive, and prescriptive paradigms.

A significant portion of healthcare data, including doctor's notes, test results, and clinical records, exists only in non-electronic, unstructured hard copies. This volume is growing rapidly. Currently, there's a push to digitize this printed data due to the rise of big data. A comprehensive exploration of big data's role in healthcare, presented in [3], proposes an architecture that leverages Hadoop-based technology. This approach integrates data from various medical levels and develops analysis methods to address medical queries effectively. For data science to effectively serve business, it is crucial to comprehend its relationship with other closely related and significant concepts. The paper [4] presents a perspective that encompasses all these concepts. It underscores data science as the connective tissue-bridging technology for

data processing (including that pertinent to "big data") and data-driven decision-making. System design with data science tools provides numerous advantages in the age of digital transformation. Using data-driven decision-making, organizations can achieve better strategies and optimized results [5]. These tools automate tasks, improving system efficiency and productivity. At the same time, predictive analytics enables the acquisition of a competitive advantage by predicting trends and risks and informing decision-making [4]. Personalization improves the user experience, drives satisfaction, and drives business growth. Risk management and fraud detection mitigate threats by detecting anomalies [6]. Innovation and product development progress with insight into market needs. Resource optimization ensures efficient allocation and cost reduction [7]. Therefore, designing systems using data science tools offer many of the challenges and benefits listed above.

System design research with data science tools is crucial for several reasons. Firstly, it aids in selecting optimal tools by evaluating functionality, scalability, performance, and compatibility with existing infrastructure, aligning tool selection with system goals. Secondly, it establishes effective methodologies for tasks like data preprocessing, model selection, and evaluation, empowering system builders and designers to create more efficient and reliable systems.

This paper outlines the framework for an advanced data science system, that integrates a research component that combines predictive analytics with epidemiological perspectives concerning the COVID-19 pandemic. Chapter 2 provides an overview of data science tools, including database systems, programming languages, machine learning libraries, and visualization tools. Chapter 3 presents a system design solution for COVID-19 data monitoring using a three-layer architecture. It covers data preprocessing, model selection, and validation methods; data layer; logic layer and the visualization layer, improving system efficiency. Deep learning techniques (LSTM) are used for predictive modeling. Chapter 4 showcases a GUI integrating LSTM modeling and SQLite for city-level COVID-19 data analysis and forecasting. Chapter 5 concludes the research, emphasizing data analysis and predictive modeling's role in enhancing decision-making and suggesting potential for further development and integration with other data sources.

II. ANALYSIS OF AVAILABLE SOLUTIONS FOR DATA SCIENCE TOOLS

When choosing a technology stack for data science, decisions must be made about the selection of elements such as database management systems, programming languages, AI/ML and visualization.

A. Database management system

SQLite, a relational database, serves as an efficient choice for data storage in small and medium-sized applications. Compared to more complex systems such as Microsoft SQL Server, PostgreSQL, or Oracle RDBMS, SQLite offers simplicity, portability, and ease of setup. It eliminates the need for a separate database server, making it suitable for projects with limited resource requirements and ease of deployment. Research [8] demonstrates that SQLite DBMS is more suitable for smaller projects. In contrast, research [9] indicates that DBMSs like PostgreSQL are more suitable for larger projects with more complex data.

B. Programming languages

Python stands out as a popular choice due to its extensive ecosystem and diverse libraries. Compared to languages like R, Julia, and SQL, Python provides a wider range of applications beyond data science. It offers integration with other technologies, making it easier to build complex systems. According to IEEE Spectrum's "The Top Programming Languages 2023" report [10], Python not only maintains its top position but further extends its dominance, particularly in AI, due to its versatile and powerful libraries, and it's also gaining ground in embedded development, making it a comprehensive choice for a wide range of developers, including beginners and professionals alike.

C. Machine learning tools

NumPy and Pandas are essential libraries for efficient data manipulation and analysis. Compared to alternatives such as MATLAB or Excel, NumPy and Pandas provide extensive functionality, better performance, and a comprehensive set of tools for working with large datasets. They enable vectorized operations and offer specialized data structures, facilitating complex data transformations and computations [11].

TensorFlow and Keras make a successful combination for building and deploying machine learning models. Compared to alternatives like Scikit-learn or PyTorch, TensorFlow and Keras stand out for their support for deep learning architectures and large-scale distributed training. They provide a high-level API, enabling researchers and practitioners to efficiently handle complex machine learning tasks.

D. Visualization tools

Matplotlib, a Python plotting library, is a contender in the field of data visualization, alongside alternatives like Tableau. While Tableau offers a simpler and more interactive interface, Matplotlib distinguishes itself with its remarkable customization and flexibility. It provides fine-grained control over visual elements and seamlessly integrates into Python code, making it well-suited for automating visualization tasks and integrating into larger workflows. However, it's important to note that Tableau comes with a significant drawback: its cost. The licensing fees associated with Tableau are considered high for small to medium-sized businesses, making it a substantial investment [12]. In contrast, other data visualization tools, including open-source options like R and Python, are available at no cost [13].

III. SYSTEM DESIGN

Designing a data science system involves creating a technology stack, typically within the realm of big data science. This stack includes components such as database systems, machine learning, and distributed systems [14]. The health system presented in [15] comprises five blocks: data collection and storage, data pre-processing, measure and concern detection, spatio-temporal information and visualization. Apache Spark is the chosen tool for this architecture, providing in-memory computations, Spark ML for machine learning, and Spark SQL for data handling.

Another example can be found in [16], where big data tools are used to monitor atmospheric composition. This system has multiple layers: data source, ingestion, Hadoop-based storage, data management, infrastructure, and monitoring/security. MySQL is used for storing the processed data, while raw Hadoop data is accessed using Python, Java, and BASH scripts.

In our own system design endeavors, we adhere to a three-layer architecture model. Our focus is on three key facets: data preprocessing, advanced AI modeling, and efficient data visualization. The decision to employ Python, SQLite, Matplotlib, NumPy, Pandas, TensorFlow, and Keras in this information system provides the system with numerous advantages compared to alternative approaches. Python's extensive ecosystem, inclusive of libraries like NumPy, Pandas, TensorFlow, and Keras, renders it versatile and widely embraced for tasks encompassing data analysis, machine learning, and general-purpose development. Our proposal for system design with data science tools is presented in Fig. 1.

A. Data collection

With the aim of designing systems using data science tools, we utilized the dataset presented in the article titled "Epidemic Location Intelligence System as a Response to the COVID-19 Outbreak in Bosnia and Herzegovina" [17]. The challenge inherent in processing this data for analytical purposes within the realm of Data Science lies in its varied forms, encompassing diverse structures,

time spans, and origins (with certain data points being recorded daily while others follow a cumulative pattern). The dataset comprises three xlsx files, and a concise overview of their contents can be found in Table 1. [18].

B. Data preprocessing

For quality analysis, uniform data is needed: by time, location and structure. When we choose a time interval, data continuity is necessary. However, for Federation of Bosnia and Herzegovina, cumulative data is given and then it seems that certain days (Sundays or holidays) are missing. That day there was not zero data, but no one entered from those responsible for the entry, so there was a sudden jump on Monday. During the analysis, it was observed that certain data had not been entered for a long period of time. For example, for Tuzla, there is no data from 11/24/2021 to 12/12/2021. We see that before that period there were 9816, and at the end 12823 cases. That difference of 3007 cases should be interpolated to empty days. Otherwise, it would be a big jump as if all 3007 new cases were on 13/12/2021. It is important to note that during data processing, the first seven instances were removed due to the lack of specified values for the municipality of Živinice.

The next step was the implementation of the function for linear interpolation. Linear interpolation is a technique for determining the function values of any intermediate point when the values of two adjacent points are known.

Linear interpolation is basically estimating an unknown value between two known values. Using the following formula, we can linearly interpolate a given data point.

$$y(x) = y_1 + (x - x_1) \cdot \frac{(y_2 - y_1)}{(x_2 - x_1)} \quad (1)$$

Here, (x_1, y_1) are the coordinates of the first data point, while (x_2, y_2) are the coordinates of the second data point, where x is the point where we interpolate and y is the interpolated value [19]. Considering the data structure within the set, interpolation was performed only on the columns that represent number of confirmed, cured cases and deaths. After the data extraction and transformation, the output data model will be represented using a series of data in the following structure, presented in Fig. 2.

C. Model creation and training

The model provides a guideline for the analysis of data on COVID-19, enabling decision-making based on evidence and understanding the spread of the virus. The model is adaptable to specific goals and challenges posed by the pandemic.

The predictive model was created using the latest deep learning technique, deep LSTM neural networks. LSTM (Long Short-Term Memory) is a model or architecture that extends the memory of recurrent neural networks. The basic form of recurrent neural networks has a problem when performing backpropagation due to its duration, which is a consequence of the large number of calculations of the hyperbolic tangent function. The goal of the LSTM model is to simplify backpropagation calculations [20]. During each computational step, the current input $x(t)$ is utilized, the previous state of short-term memory $c(t-1)$, and the prior state of the hidden state $h(t-1)$, which is shown in Fig. 3 [21].

When assessing the model, the Mean Squared Error (MSE) quantifies the average squared variance between predicted and actual values. The MSE of 655.7611 highlights significant discrepancies due to magnified errors from squared deviations.

The Root Mean Square Error (RMSE) complements MSE, offering a more understandable measure of predic-

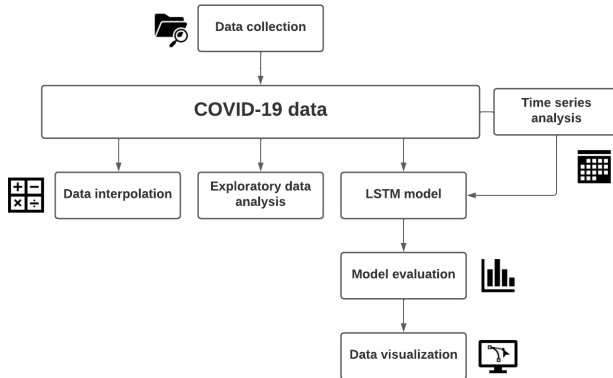


Fig. 1: Algorithms and processes used in this research.

TABLE I: Available files in the dataset.

The name of the file	The contents of the file
zzjz_opcine_fbih.xlsx	Contains information about the name of the municipality, the ID of the municipality, dates, the total number of confirmed, cured and deaths
zzjz_opcine_rs.xlsx	Contains information about the name of the municipality, the ID of the municipality, dates, the daily number of confirmed cases and deaths
zzjz_brcko.xlsx	Contains information about the name of the municipality, the ID of the municipality, dates, number of tested, positive, cured, dead, active, hospitalized, seriously ill and daily confirmed

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['Tuzla' 9346 8943 348 Timestamp('2021-05-28 00:00:00')]
['Tuzla' 9350 8949 348 Timestamp('2021-05-29 00:00:00')]
['Tuzla' 9354 8955 348 Timestamp('2021-05-30 00:00:00')]
['Tuzla' 9358 8961 348 Timestamp('2021-05-31 00:00:00')]
['Tuzla' 9362 8967 348 Timestamp('2021-06-01 00:00:00')]
['Tuzla' 9366 8973 348 Timestamp('2021-06-02 00:00:00')]
['Tuzla' 9371 8979 349 Timestamp('2021-06-03 00:00:00')]
['Tuzla' 9375 8989 351 Timestamp('2021-06-04 00:00:00')]
['Tuzla' 9378 8990 351 Timestamp('2021-06-05 00:00:00')]
['Tuzla' 9381 8992 351 Timestamp('2021-06-06 00:00:00')]
['Tuzla' 9385 8994 351 Timestamp('2021-06-07 00:00:00')]
['Tuzla' 9386 8999 351 Timestamp('2021-06-08 00:00:00')]
['Tuzla' 9387 9005 351 Timestamp('2021-06-09 00:00:00')]
['Tuzla' 9392 9005 351 Timestamp('2021-06-10 00:00:00')]
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Fig. 2: View of the output model.

tive inaccuracies. Derived by taking the square root of MSE using np.sqrt , RMSE restores the original variable scale. An RMSE of 25,607 implies an average prediction error of about 25,607 units. Within the range of values (9319-18109), this RMSE suggests reasonable alignment of model predictions with authentic values.

Apart from the metrics mentioned above, visualizing the predicted and actual values can offer a graphical depiction of the model's efficacy in capturing data patterns and trends. It can also aid in pinpointing discrepancies or possible overlaps when notable deviations exist between predicted and actual values. Fig. 4 illustrates the training and validation loss ratio, serving the purpose of identifying potential instances of overfitting.

D. Visualization component

The visualization component in system design with data science tools is a pivotal aspect that bridges the gap between complex data and human understanding. It empowers data scientists, analysts, and decision-makers to extract valuable insights from data, make informed choices, and drive innovation across various domains. Here's an overview of the visualization component within the context of our system design with data science tools:

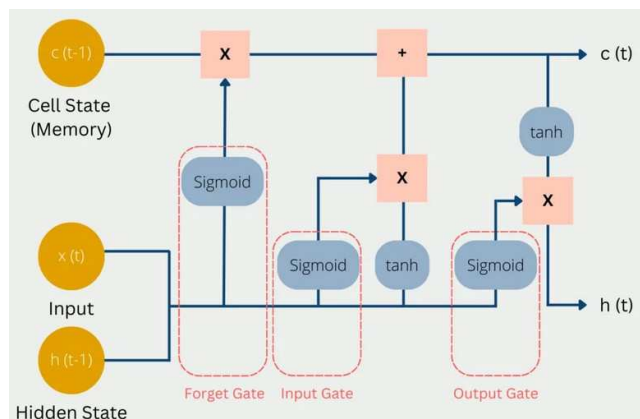


Fig. 3: LSTM architecture.

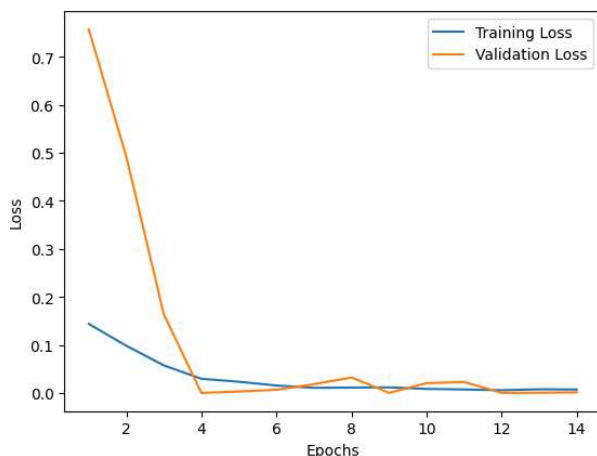


Fig. 4: Loss ratio during training and validation.

- Data representation: This involves selecting appropriate graphs or dashboards that effectively convey information.
- Exploratory data analysis (EDA): During the data exploration phase, data scientists use visualization to uncover patterns, trends, outliers, and correlations within the dataset.
- Model evaluation: In machine learning and data modeling, visualization aids in evaluating model performance. These visuals help in selecting the best model for a specific task.
- Dashboard design: Dashboards are comprehensive visual interfaces that provide an overview of key performance indicators (KPIs) and insights.
- Customization: Tailoring visualizations to specific user requirements and domain needs is essential. Customization options should allow users to choose color schemes, labels, and visualization types that align with their objectives.

The visualization results are presented in the next chapter.

IV. RESULT ANALYSIS

Our software works as a generalized platform that is highly customizable, so the user can easily see predictive modeling using Matplotlib. This allows decision-makers to analyze the data to visualize trends. The analysis of the solution contains a display of the graphical user interface, which includes the integration of LSTM predictive modeling and the SQLite database. This GUI provides a framework for analyzing and forecasting COVID-19 data at the individual city level.

The "covid19predict.db" database stores COVID-19 prediction data for multiple cities in the "citydata" table. This table contains predicted total confirmed, recovered, and deceased cases for specific dates, generated by an LSTM model trained on historical data.

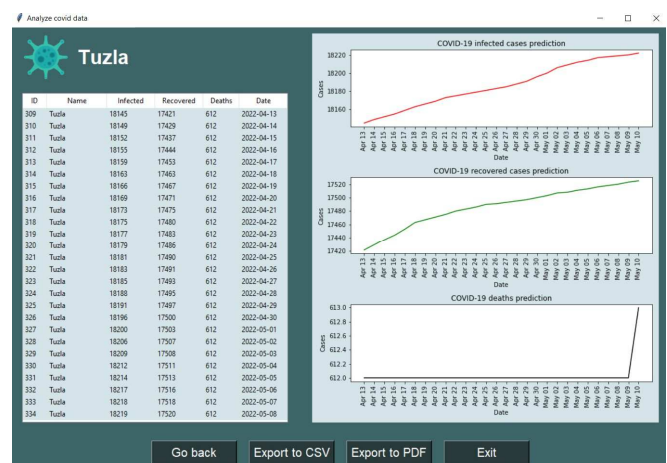


Fig. 5: User interface with tabular data display and line graphs for Tuzla.

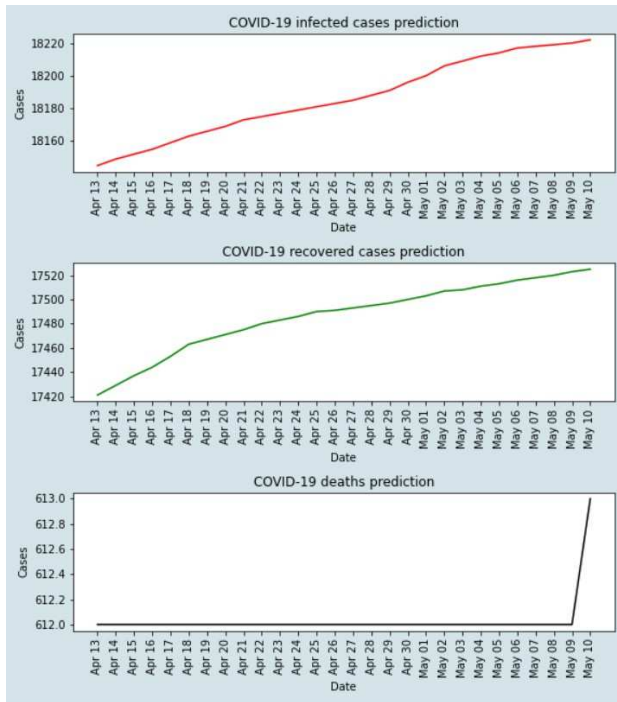


Fig. 6: Display of line graphs (trends of infected, recovered and dead over time) for Tuzla.

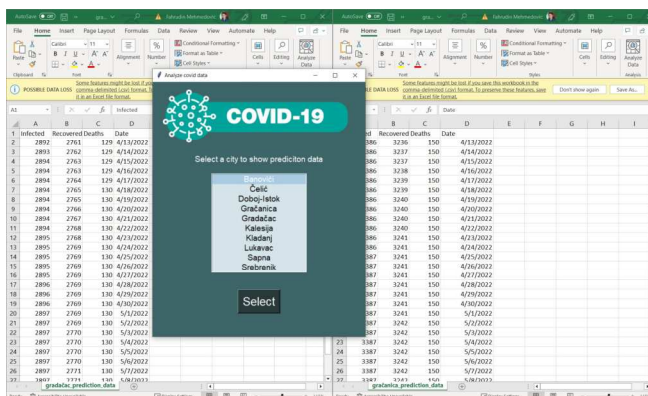


Fig. 7: Display of the user interface with created CSV reports for the cities of Gradacac and Gračanica.

The downloaded data is displayed in a table format using the Treeview widget, showing information like ID, city name, infected, recovered, and deceased counts, and corresponding dates. This table is depicted in Fig 5. Additionally, the program generates line plots with Matplotlib to visualize trends in infections, recoveries, and deaths over time. The x-axis represents dates, and the y-axis shows the case numbers. These trends are illustrated in Fig. 6.

Apart from data visualization, the program provides various functionalities for exporting and preserving data for subsequent analysis, as illustrated in Fig. 7. Users have the option to export forecast data for the chosen city to a CSV file. The program retains data in CSV format, facilitating effortless importation into spreadsheet applications or other data analysis tools.

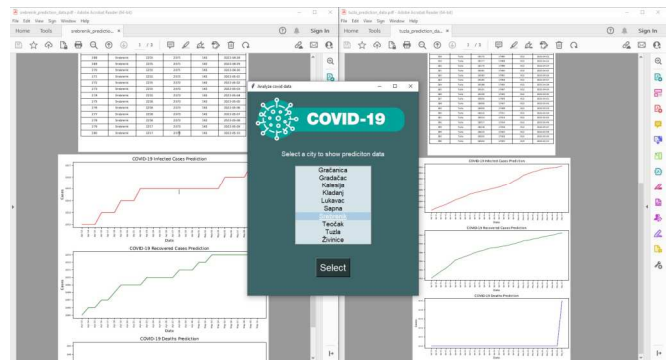


Fig. 8: Display of the user interface with created PDF reports for the cities of Srebrenik and Tuzla.

Moreover, users can generate a comprehensive report in PDF format by clicking the "Export to PDF" button. The report includes a title, a paragraph describing the report's purpose, forecast data presented in tabular form, and graphics illustrating trends. The resulting PDF report offers a concise and authoritative assessment of the COVID-19 situation in the chosen city, which facilitates decision-making and the development of response strategies. The details of the report layout and contents can be found in Fig. 8.

V. CONCLUSION

The paper contributes significantly to digital transformation through data science, focusing on predictive analytics and computer architecture in healthcare AI within Industry 4.0. The proposed three-layer information system architecture monitors COVID-19 epidemiological data using LSTM deep neural networks, improving system efficiency and data quality. It integrates LSTM modeling with an SQLite database for city-level COVID-19 analysis. The technology stack's advantages are highlighted.

The research process involves defining the COVID-19 prediction problem, data collection and preprocessing, feature engineering, model selection, and implementation with visualization. Future iterations can include real-time data integration and prescriptive analytics.

The paper emphasizes the comprehensive approach to system design with data science tools and the importance of collaboration with domain experts. It underscores the significance of integrating data science in healthcare and Industry 4.0 for optimized processes, better patient outcomes, and resource allocation.

In conclusion, this research showcases the transformative potential of data analysis and predictive modeling, driving digital transformation, informed decision-making, and continuous development across domains. Integrating data science tools in information systems enhances effectiveness and empowers organizations in the data-driven era.

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